

5(a)(i)	correct answer as a whole number 34(%) ;	1
5(a)(ii)	allow enzyme for trypsin allow substrate for casein any five from: 1 at lower concentrations there are not enough, active sites / enzymes ; 2 idea of substrate molecules cannot enter active site until product released ; 3 as concentration increases, increased number of active sites ; 4 idea that as concentration increases there are more successful collisions between enzyme and, substrate / casein ; 5 (so) allows increase in (number of) enzyme-substrate complexes (formed per unit time) ; 6 increased rate of casein breakdown as enzyme concentration increases ; 7 <i>ref. to</i> casein reaction e.g. milk becomes transparent as solution contains amino acids ; 8 AVP ; e.g. depends on how many active sites available at any one time <i>ref. to</i> enzyme-substrate complex formation per unit time at high concentration substrate concentration is becoming the limiting factor when clear, total hydrolysis of casein has occurred	5

Question	Answer	Marks
5(b)(i)	any one from: immobilising the enzyme may have altered the, tertiary structure of the enzyme / shape of the active site ; <i>suggestion that</i> material may not be inert and may have an inhibitory effect at lower temperatures ; immobilising may have covered, part of the active site / active site of some enzymes ; <i>idea that</i> trypsin free in solution has increased chance of collision with substrate ; immobilising changes the optimum temperature of the enzyme ;	1
5(b)(ii)	any two from: immobilisation, protects / stabilises / AW, the, enzyme / trypsin ; from (thermal) denaturation ; ora enzyme free in solution denatured maintains shape of active site ; immobilisation may result in lower, kinetic energy / vibration, of the enzyme molecules compared to free (so protects from denaturation) ; ora prevents the breaking of bonds holding the tertiary structure (of immobilised enzyme molecule) ; ora <i>ref. to</i>, hydrogen / ionic bonds ;	2